

Avinash Chauhan

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Academic Profile

Undergraduate Researcher at IISER TVM specializing in the intersection of **Stochastic Processes, Optimization, and Reinforcement Learning**. Leveraging strong mathematical foundations for **High-Frequency Trading (HFT)** modeling and **Quantum Machine Learning** research.

Education

IISER Thiruvananthapuram

BS-MS in Mathematics (Major: *i2 Mathematics*); GPA: 7.21/10.0

Kerala, India

Aug 2023 – May 2028

– **Honors:** PM Yashasvi Scholar for Academic Excellence | **GATE 2026 (DA): AIR 3144.**

– **Coursework:** Numerical Analysis, ODEs, Probability, Linear Algebra, Optimization.

Gyan Deep Sr. Sec. Public School

Higher Secondary Education (Science); Percentage: 95.4%

Uttar Pradesh

Aug 2020 – May 2022

Technical Arsenal

Languages: Python, SQL, R, C++, $\text{\LaTeX}$

ML Frameworks: PennyLane, PyTorch, TensorFlow, Keras, Scikit-Learn, Pandas, NumPy, SciPy

Specializations: Deep RL (SAC), Quantum ML (VQC), CNNs, Transformers, PCA, K-Means

Tools: Git, GitHub, Jupyter Notebooks, VS Code

Core Research Experience

Research Intern | WSAI, IIT Madras

Summer 2026

- Executed computational research under the supervision of **Prof. Karthik Raman** on the project titled **"Discerning ecological behaviour and interactions of minimal reactomes of the human gut."**.
- Modeled the human gut microbiome mathematically, bridging systems biology with large-scale computational optimization frameworks..

Reinforcement Learning for Market Making | Python, PyTorch, Gym

Feb 2026 – Mar 2026

- Engineered a **Soft Actor-Critic (SAC)** agent to optimize bid-ask spreads in a simulated Limit Order Book (LOB).
- Modeled inventory risk using asymmetric reward functions to manage stochastic price volatility.
- Benchmarked performance against classical Avellaneda-Stoikov strategies, achieving superior risk-adjusted returns.
- [View Source Code](#)

Classical-Quantum Hybrid Classifier | Python, PennyLane, Quantum ML

Oct 2024 – Nov 2024

- Developed a novel hybrid protocol utilizing PCA and K-Means for reference centroid selection.
- Implemented a $UU^\dagger$ quantum circuit to measure state overlap probability for non-linear classification.
- Optimized decision rules for near-term quantum hardware by leveraging Hilbert space embedding.
- [View Source Code](#)

Game Theory Reading Project | IISER TVM, Supervisor: Prof. Sheetal

Dec 2025 – Jan 2026

- Analyzed strategic decision-making and Nash equilibria in multi-agent competitive environments.
- Explored payoff matrices and mathematical modeling of strategic interactions in economic contexts.
- [View Project Report](#)

Selected Mathematical Implementations

Prey Population Dynamics (Lotka-Volterra Model) | Python, SciPy

Independent

- Simulated predator-prey stability patterns by numerically solving systems of Lotka-Volterra ODEs.
- Validated simulation results against historical Hudson Bay datasets to analyze oscillatory behavior.
- [View Source Code](#)

Selected Technical Projects

Malaria Parasite Detection using CNNs <i>Python, TensorFlow</i>	Independent
– Built a CNN to automate malaria detection from microscopic blood cell images with high accuracy. – Demonstrated the application of deep learning in healthcare analytics through data preprocessing and hyperparameter tuning. –  View Source Code	
Customer Segmentation Analysis <i>Python, SQL, Scikit-Learn</i>	Independent
– Pre-processed transaction data and applied PCA and K-Means for high-value customer archetype identification. –  View Source Code	

Certifications & Development

Mastering Deep Learning IIM Bangalore (SWAYAM)	Jan 2025
Intl. Conference on Computational & Applied Math	Dec 2025
AI & ML Workshops Deloitte, WEKA, IIT Madras	2024 – 2025